

SELO YARTSEVO, Russian Federation

WMO: 239870

Lat: 60.2533N

Lon: 90.2019E

Elev: 58

StdP: 100.63

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.2	(c) -41.7	(d) -47.3	(e) 0.0	(f) -45.0	(g) -44.2	(h) 0.0	(i) -41.8	(j) 8.8	(k) -13.9	(l) 8.2	(m) -14.9	(n) 0.6	(o) 90	(p) 0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.7	(c) 28.6	(d) 19.1	(e) 26.6	(f) 18.2	(g) 24.6	(h) 17.5	(i) 20.6	(j) 26.0	(k) 19.5	(l) 24.6	(m) 18.5	(n) 23.1	(o) 2.6	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	23.0	17.6	12.7	21.9	16.5	11.9	20.8	59.8	26.0	56.1	24.7	52.7	23.1	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.2	(b) 7.2	(c) 6.3	(e) -45.6	(f) 32.1	(g) 3.8	(h) 2.5	(i) -48.3	(j) 33.9	(k) -50.5	(l) 35.4	(m) -52.7	(n) 36.8	(o) -55.4	(p) 38.6		
			DB			3.7	1.5	-48.3	23.5	-50.5	24.4	-52.6	25.3	-55.3	26.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	-23.4	-19.4	-9.7	-1.4	6.5	15.6	18.2	14.4	7.8	-1.1	-12.9	-19.3
	DBStd	15.77	10.77	8.82	7.21	5.79	5.36	4.80	3.49	3.81	4.34	5.64	9.21	10.08
	HDD10.0	4999	1037	823	610	342	137	11	0	7	94	344	687	909
	HDD18.3	7486	1295	1056	868	592	367	107	46	131	318	602	937	1167
	CDD10.0	632	0	0	0	1	29	178	253	144	27	1	0	0
	CDD18.3	76	0	0	0	0	1	24	41	10	0	0	0	0
	CDH23.3	700	0	0	0	0	16	259	344	79	2	0	0	0
	CDH26.7	163	0	0	0	0	3	62	84	14	0	0	0	0
	WSAvg	2.9	2.8	2.7	3.1	3.3	3.2	2.6	2.3	2.5	2.8	3.2	3.1	3.1
(15) Precipitation	PrecAvg	612	38	33	29	32	47	61	62	72	69	65	55	49
	PrecMax	827	66	47	61	74	107	109	113	163	127	116	101	80
	PrecMin	423	3	12	10	8	17	5	2	13	15	25	27	14
	PrecStd	97	15	9	13	16	22	26	29	32	26	19	18	18
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.2	-0.6	6.5	14.9	25.4	31.0	31.6	28.9	22.5	13.0	2.3	-0.7
		MCWB	-3.0	-1.2	2.1	7.6	15.2	19.2	20.7	20.2	15.0	8.3	1.5	-1.1
	2%	DB	-4.4	-2.7	3.5	11.4	21.7	28.5	29.0	25.6	19.2	9.9	1.2	-2.5
		MCWB	-5.0	-3.4	0.4	5.7	12.8	18.2	19.8	18.6	13.7	7.6	0.5	-3.1
	5%	DB	-7.0	-4.9	1.6	8.7	18.8	26.5	26.9	23.2	16.6	8.0	0.3	-4.3
		MCWB	-7.5	-5.4	-0.5	4.3	11.7	17.3	19.0	17.3	12.6	6.0	-0.2	-4.7
	10%	DB	-9.3	-7.6	0.1	6.4	15.5	24.2	25.1	21.0	14.3	6.1	-0.8	-6.6
		MCWB	-9.7	-8.1	-1.4	2.7	9.7	16.6	18.2	16.6	11.6	4.7	-1.4	-7.0
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-1.2	2.6	8.6	16.0	20.7	22.7	21.2	16.2	9.7	1.8	-1.0
		MCDB	-2.2	-0.9	4.9	13.5	23.8	27.7	28.3	26.5	20.7	12.2	2.1	-0.8
	2%	WB	-4.9	-3.4	0.9	6.2	13.8	19.4	21.1	19.6	14.5	7.9	0.7	-3.0
		MCDB	-4.2	-2.8	2.7	10.6	20.2	26.0	26.5	24.4	18.1	9.6	1.1	-2.5
	5%	WB	-7.4	-5.5	-0.2	4.6	12.2	18.4	20.1	18.1	13.1	6.3	-0.1	-4.7
		MCDB	-7.0	-4.9	1.5	8.1	17.8	24.4	25.2	21.7	15.8	7.8	0.3	-4.3
	10%	WB	-9.7	-8.1	-1.5	3.2	10.3	17.4	19.1	16.8	11.8	4.7	-1.4	-7.0
		MCDB	-9.2	-7.6	0.1	6.0	15.1	22.7	23.6	20.1	14.0	6.0	-0.8	-6.6
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	9.1	10.6	9.5	9.7	10.2	9.7	8.5	7.3	5.4	6.5	6.9
		MCDBR	7.6	8.3	10.0	12.2	14.7	14.6	13.3	12.6	11.8	7.7	4.1	6.3
		MCWBR	7.4	8.1	7.9	7.7	7.8	6.3	5.7	6.3	6.8	5.5	4.2	6.3
		MCDBR	7.6	8.3	8.4	10.7	13.4	12.1	11.0	10.4	9.8	6.9	4.0	6.4
	5% WB	MCWBR	7.4	8.1	7.0	7.4	7.6	5.8	5.7	5.8	6.1	5.2	4.2	6.4
(40) Clear-Sky Solar Irradiance	taub		0.272	0.308	0.342	0.368	0.386	0.394	0.453	0.407	0.359	0.326	0.272	0.251
	taud		1.941	1.975	1.983	2.050	2.253	2.371	2.176	2.297	2.395	2.252	1.995	1.894
	Ebn at Noon		433	642	755	816	837	837	767	778	759	641	430	297
	Edh at Noon		50	88	121	136	121	110	130	107	82	67	46	34
(44) All-Sky Solar Radiation	RadAvg		0.44	1.33	2.67	4.16	4.86	5.53	5.11	3.65	2.18	0.97	0.47	0.22
	RadStd		0.06	0.12	0.30	0.42	0.52	0.67	0.38	0.33	0.33	0.14	0.10	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page